

FLCR-09 - Lattice Closure And Terminal Addresses

Series: Formalizing LCR, Unifying Standard Models **Artifact:** formal paper source **Status:** first-pass rich rewrite; requires final citation and build pass. **Claim posture:** maximal local claim posture with explicit lane boundaries.

Abstract

This paper installs the lattice and code ladder as an addressable terminal surface. It distinguishes construction claims, lookup claims, invariant claims, glue claims, and physical-mapping claims. The strongest local result is that lattice-forge and CAM surfaces can provide no-cost lookup routes for E8, Niemeier, Leech-facing, and related terminal addresses once receipts are attached.

Keywords

E8; Leech; Niemeier; Golay; lattice lookup.

External Reader Orientation

An outside reviewer should read this paper as a formal-system construction paper. Its local object is **Lattice Closure And Terminal Addresses**. The paper's immediate contribution is: Binds code/lattice ladders, no-cost lookup surfaces, and invariant boundaries.

The nearest external anchors are finite-state reasoning, typed interfaces, proof-carrying records, conservative extension, and ordinary mathematical citation discipline. LCR terms are therefore internal labels for the construction unless a row explicitly imports an external theorem, citation, dataset, or calibration target.

It does not ask the reader to accept any Standard Model or literal-physics identification at this layer. Such mappings are deferred to the translation papers and must carry their own evidence.

Reviewer Compression

Core object. Lattice Closure And Terminal Addresses.

Primary result. This paper contributes the local formal object and its safe downstream-use rule.

Primary non-result. This paper does not claim direct identity with Standard Model or physical objects.

Review strategy. Evaluate the formal claim rows first, then inspect the receipt/citation/calibration rows that support each claim. Appendices provide routing and implementation context; they should not be read as stronger than their lane permits.

Peer Reviewer Assessment

Reviewer assumption: the reviewer has been briefed on the FLCR structure, claim lanes, CAM/crystal routing, forge/action surfaces, and the distinction between internal formal results and external physical calibration.

Recommendation. major revision, technically promising.

Review score vector. orientation=2, claim_granularity=2, evidence_visibility=2, falsifier_visibility=2, journal_apparatus=1, base_layer_boundary=2.

Formal claim rows reviewed. 3.

Reviewer Strength Finding

The manuscript is now readable as a bounded formal-system paper: it identifies its local object, separates internal construction from physical interpretation, and exposes claim lanes for review.

Major Revision Requests

- Convert the strongest proof sketch into numbered definitions, lemmas, and propositions with explicit dependencies.
- Replace placeholder source classes with final citation keys, receipt hashes, or dataset identifiers wherever possible.

- Move repeated pass-derived material into a compact evidence appendix and keep the main body focused on the paper-local argument.
- Add at least one worked example showing the local object, legal operation, receipt, residue, and downstream import rule.

Minor Revision Requests

- Normalize theorem capitalization and punctuation.
- Add final figure/table captions where the text currently describes diagrams schematically.
- Ensure every acronym and local term appears in the paper-local glossary.

Acceptance Blockers

- A reviewer can accept the formal contribution without accepting later physics claims, but only if final citations and receipt identifiers are attached.
- The proof sketch must be promoted into a formal proof or explicitly labeled as a construction argument.

1. Contribution And Scope

- Places the finite LCR stack on a code/lattice ladder.
- Separates lookup capability from full invariant proof.
- Defines terminal lattice addresses as CAM-queryable objects.
- Routes nontrivial glue cosets, Gamma72, and physical maps to explicit downstream receipts.

This paper belongs to the LCR-native construction band. Physics and Standard Model language, when it appears, is only a deferred mapping cue, analogy, or explicitly bounded calibration target. The paper's base object must remain stable enough for later papers to cite without importing a stronger physical claim by implication.

2. Source Routing

Primary routed sources:

- 08_Lattice_Closure.md
- supplements/08_INTERNAL_REASONING_SUPPLEMENT.md

Cross-corpus evidence classes:

- CAM_CLAIM_100_SCORE_AUDIT.md
- NSIT_TOOL_CLOSURE_MATRIX.md
- NSIT_REASONED_CLOSURE_PASS.md
- PAPER_REWORKS_CRYSTAL_PROJECTION.json
- standards, glue, window, and node databases where applicable
- notebook-derived review prompts and orbital source manifests

3. Definitions

- **Construction claim.** A claim that a code or lattice object is generated by a declared construction.
- **Lookup claim.** A claim that a library or CAM surface can retrieve an already-addressed object without re-deriving all invariants.
- **Glue claim.** A claim about discriminant-form or coset data needed for overlattice construction.
- **Terminal address.** A stable lattice/CAM address consumed by later papers as a lookup handle.

4. Formal Claims

Claim	Lane	Statement
Theorem 9.1	cam_crystal_reapplication_result	The FLCR lattice ladder supports receipt-bound terminal lookup across the declared code/lattice chain.
Proposition 9.2	receipt_bound_internal_result	No-cost Leech lookup is a valid computational capability once the lookup receipt is attached, but it is not a proof of every Leech invariant.
Protocol 9.3	falsifier_or_open_obligation	Nontrivial glue, Gamma72, orbit, and physical-map claims require their own evidence lanes.

Reviewer Claim Ledger

This ledger expands the formal-claims table into review actions. It is intended to make proof granularity explicit: what is being claimed, what evidence type can support it, what boundary remains, and what the next review action is.

Formal row	Type	Claim in review terms	Evidence required	Boundary	Next review action
Theorem 9.1	CAM route	The FLCR lattice ladder supports receipt-bound terminal lookup across the declared code/lattice chain	CAM/crystal route, source node, projection, window, or addressable evidence artifact	routing and reapplication evidence; not a standalone physical proof	check the CAM source, route, and attached data are named
Proposition 9.2	finite/executable result	No-cost Leech lookup is a valid computational capability once the lookup receipt is attached, but it is not a proof of every Leech invariant	receipt, validator, executable pass, generated artifact, or finite enumeration	the stated finite/executable domain only	verify the receipt exists and its scope matches the claim
Protocol 9.3	obligation/falsifier	Nontrivial glue, Gamma72, orbit, and physical-map claims require their own evidence lanes	explicit missing derivation, adapter, receipt, dataset, comparator, or failed test condition	does not negate satisfied lower-level rows	name the next binding action and owner surface

Claim Granularity And Review Table

The table below separates the claim types so the paper is reviewable without accepting the whole FLCR vocabulary at once.

Claim type	What this paper may claim	Acceptance test	What is not claimed by that row
Formal-system result	FLCR-09 defines or uses Lattice Closure And Terminal Addresses as a typed FLCR object with declared inputs, operations, outputs, and residue.	Definitions, formal claims, construction steps, and downstream-use rules are internally consistent and lane typed.	External physical truth, measured prediction, or identity with a standard theory.
Computational or receipt-bound result	Enumerations, TarPit runs, CAM/crystal routes, verifier rows, and generated artifacts are claims about the stated finite or executable domain.	A named receipt, validator, manifest, pass report, or rebuild result exists and matches the row scope.	Completeness outside the enumerated domain or physical correctness outside the tested comparator.
Imported theorem or external definition	Imported mathematics remains an external theorem/citation target; this paper may use it only through declared mappings and conservative-extension discipline.	The source theorem, definition layer, or citation target is named and the mapping into this paper is explicit.	A new proof of the imported theorem or a hidden change in the imported object's meaning.
Interpretive bridge or analogy	Analog, workbook, visual, or translation language may explain why the construction is useful.	The analogy preserves the relevant structure and does not promote itself into a theorem.	That the analogy alone proves the formal, computational, or physical claim.
Physical or calibration-facing claim	Physical or Standard Model identity is not claimed here unless the paper contains an explicit calibration row; the default status is deferred mapping.	A dataset, citation, calibration protocol, uncertainty, comparator, or falsifier is attached.	A physical conclusion supported only by shared vocabulary rather than calibration, comparator data, or a stated falsifier.
Open obligation or falsifier	A missing derivation, adapter, receipt, dataset, or failed comparator is a named research channel.	The next binding action and the reason the stronger claim is not closed are stated.	That the base formal result is false merely because a stronger downstream claim remains unfinished.

5. Paper-Specific Development

1. Classify each lattice statement as construction, lookup, invariant, glue, or physical-map claim.
2. Bind terminal addresses to receipt-bearing library or CAM lookup surfaces.

- 3. Allow no-cost lookup only for the addressed object and attached receipt scope.
- 4. Route nontrivial glue, Gamma72, orbit, and physical interpretation claims to separate evidence lanes.

Proof Sketch

The key argument is burden separation. A lookup receipt proves that the system can retrieve or address the declared object under the declared library surface. It does not automatically prove every invariant of that object. This separation lets later papers use Leech/E8-style capability immediately while preserving the stronger mathematical burdens.

6. Construction

The construction is intentionally staged. First, the paper names the finite or typed object that can be inspected directly. Second, it declares the admissible operations over that object. Third, it records the receipt or source class that allows another paper to consume the result. Fourth, it names the residue that must not be silently promoted.

For this paper, the operative object is Lattice Closure And Terminal Addresses. Its safe consumption rule is:

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source object -> declared operation -> receipt/lane -> downstream import
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The unsafe consumption rule is:

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source phrase -> downstream identity claim
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That second pattern is explicitly rejected by FLCR-01 and must be rewritten as a claim-lane promotion with evidence.

7. Evidence And Receipts

Evidence source	What it contributes
Paper 08 legacy body	Lattice closure, E8/Niemeier/Leech-facing chain, terminal addresses.
Internal supplement 08	Construction A ladder, Hamming/Fano, Golay/Leech, glue-code discipline.
NSIT tool matrix	E8 root enumeration, lattice code chain, Niemeier:E8^3 carrier, F2 bridge and registry receipts.
CAM Claim 100 Score Audit	Paper 08 scored 91 as system-composition closeable.

Appendix A. Recursive Capability Reapplication Review

This review treats the newly admitted capability families as operators. The question is not only what each source says, but what it solves when the source is recursively reapplied through CAM, claim lanes, forge admission, receipt loops, and paper-to-paper routing.

Operator Effects

Operator	Standalone result	Recursive result	Newly resolved state	Remaining boundary
REAPPLY-GLM-OBLIGATION-CLOSURE	The GLM closure matrix independently compares verifier findings to named obligations and advances 11 obligations to partial closure while identifying the untouched set.	When fed back through the claim lanes, the matrix stops the suite from carrying stale open labels where bounded verifier evidence already exists, while preserving untouched obligations as active next channels.	Obligation status promotion becomes auditable instead of rhetorical.; Papers 09, 10, 13, 16, 18, and 33 gain bounded closure support from specific verifier findings.; The suite gains a clean split between partial closure, engineering scope, and untouched physical/calibration bridges.	Partial closure is not unconditional closure.; CKM, quark/color measurement, Higgs/QFT mass calibration, GR curvature bridges, and full Moonshine remain separate dependency lanes where the matrix says they remain open.

The practical consequence is that several prior gaps are now better classified. Some are closed as routing, admission, finite-address, or executable-publication problems. The residues that remain are sharper: exhaustive source intake, physical calibration, full paper-local runner parity, external measurement, and domain-specific validation.

Appendix B. Broad Capability Intake

This addendum admits non-obvious source material by function rather than filename. Each row states what the source now lets the paper do, while detailed receipt provenance remains in the evidence report instead of the formal claim body.

Capability	Lane	Paper effect	Boundary
CAP-GLM-OBLIGATION-CLOSURE	normal_form_result	Adds a closure matrix that compares independent GLM verifier findings against open obligations and records bounded promotions instead of leaving the papers in their earlier unresolved posture. Evidence facts: 12 findings considered against 29 obligations; 11 obligations advanced to partial; 17 untouched; closed-bounded findings include GLM-F1-SM01, GLM-F2-SP002, GLM-F4-RULE30P3, GLM-F6-YM-EXT02, GLM-F7-SM04, GLM-F8-HODGE-EXT01.	Promotions are lane-bound and partial where the matrix says partial; external physical bridges remain separate calibration obligations.

The resulting claim posture is broader but still auditable: a paper can import a capability when the evidence lane is satisfied, and any remaining gap is named as an adapter, calibration, rerun, or falsifier obligation.

Appendix C. Implementation Intake

This addendum binds implementation work that was not fully represented in the earlier paper routing. The rows below are not pasted source text; they are evidence-lane imports that change what the paper can claim now.

Implementation source	Lane	Claim effect	Boundary
IMPL-E8-LEECH-RUNTIME-WITNESS	receipt_bound_internal_result	Adds concrete runtime witnesses: E8 neighbor graph with 168 nodes-with-hits and 6872 edges, plus Leech24 octad witness with 256 nodes and 27264 edges under an ok ledger. Evidence facts: lattice_type=E8; nodes_considered=168; nodes_with_hits=168; edges_added=6872; relation=e8_kiss; lattice_type=LEECH24_WITNESS; nodes_added=256; edges_added=27264.	Closes witness construction and receipt integrity for these runs; full uniqueness or classification claims require cited lattice theory and reproducible rerun protocol.

The immediate paper effect is stronger claim posture where these implementation rows satisfy the relevant lane. Remaining caution is reserved for specific claims that still lack an adapter, comparator, calibration target, or rerun receipt.

Appendix D. Resolved-State Closure Pass

This pass removes false restrictions on the paper's claim posture. A row is no longer called open merely because a stronger future claim exists. The satisfied lane is closed now; only the stronger claim that lacks its required adapter, comparator, calibration, or falsifier remains as residue.

Closed Now

Row	Lane	Resolved state	Remaining boundary
paper lift enumeration	receipt_bound_internal_result	8/8 LCR states succeeded; error walls=0; avg TarPit mass=0.505787.	This closes the paper-local finite lift readout, not every stronger physical interpretation.
terminal addressability	normal_form_result	The lattice/terminal surface is closed as an addressable lookup and invariant/residue split.	The family closure is a structural lane; measured physics still requires destination-specific calibration.
decade-1 tower	receipt_bound_internal_result	The decade tower is resolved with avg TarPit mass=0.510370 and mass sum=40.829567.	The decade total is an internal tower metric, not a standalone universal physical constant.
family-09 cross-lift comparison	cam_crystal_reapplication_result	Family tower binds FLCR-09, FLCR-19, FLCR-29, FLCR-39; avg TarPit mass=0.505916; error walls=0.	Cross-lift agreement strengthens routing and comparison; it does not erase paper-local boundaries.
terminal address surface	cam_crystal_reapplication_result	Lattice terminal addressability is closed as lookup and invariant split.	Uniqueness of physical landing forms remains receipt-bound.

Claim Posture After This Pass

FLCR-09 should now be read as a resolved-state contribution for Lattice Closure And Terminal Addresses at its declared lane. The paper may state the strongest claim supported by these rows directly. It should reserve caution only for a specifically named stronger claim whose required evidence is absent.

8. Dependencies And Downstream Use

This paper may be imported by later FLCR papers only through the claim lanes above. The default downstream consumers are:

- FLCR-11 through FLCR-18 for window, receipt, admission, CA, algebraic, curvature, residue, and exceptional machinery.
- FLCR-28 through FLCR-30 for CAM/crystal, corpus ribbon, and universal normal-form intake.
- FLCR-31 through FLCR-40 only after the LCR-native result has been cited and the translation claim has its own evidence lane.

9. Limitations And Falsifiers

- Lookup capability is not identical to proof of all automorphism, orbit, or invariant data.
- Identity glue evidence must not be promoted as nontrivial glue evidence.
- Lattice addresses do not supply physical units or empirical identity by themselves.

A falsifier for this paper is any example that satisfies the declared input conditions while violating the stated construction, receipt, or lane boundary. An interpretation that merely wants a stronger downstream conclusion is not a falsifier; it is an obligation routed to a later paper.

10. Publication Readiness Tasks

- Validator identities and receipt hashes are recorded in the manifest or receipt appendix.
- External theorems require final citation entries before journal submission.
- Every theorem, proposition, and protocol must retain a claim lane and evidence boundary.
- The Markdown, LaTeX, and PDF forms are generated from the same canonical source.

Dual Positioning: Story And Formal Carrier

This paper must be read in two synchronized ways. Sequentially, it explains why the next state exists in the corpus story. Formally, it binds that state to accepted mathematics, IRL formalism, code receipts, validators, CAM/crystal lookups, and claim-lane boundaries.

Assigned original ribbon role(s): 08/terminal_action.

Original slot	Family	Lift	Role	Proof form
08	terminal_action	0	order-1 slot-8: land into lattice, game, runtime, or terminal address surfaces	terminal lookup, invariant split, and addressability proof

The formal obligation is to state the strongest valid claim available for this paper without letting interpretation silently change the addressed object. Any author interpretation belongs beside the formalism as a declared relabeling, bridge, or analog, and must be accompanied by the evidence lane that makes the claim consumable by later papers.

State-Bound Proof Contract

This paper receives: state emitted by prior slot 07 and reopened at original slot 08 (Lattice Closure).
It must produce: formal result of original slot 08 plus the same-family lift path toward slot 18.

Semantic transition: land the state in a terminal lattice, game, runtime, or address surface.

Accepted formalism targets to bind in the journal rewrite:

- lattice theory
- finite-state systems
- terminal object/addressing templates
- lookup-table construction and invariant checks

Slot	Family	Lift	Produced proof form
08	terminal_action	0	terminal lookup, invariant split, and addressability proof

The formal carrier uses these accepted tools while preserving interpretive claims as explicit relabelings, correspondences, or bridges. Claim promotion is admissible only when a receipt, standard citation, CAM/crystal reapplication, normal-form result, calibration datum, or falsifier boundary is attached.

Provenance And Crystal Evidence Integration

This section integrates prior work, CAM/crystal projections, NSIT tooling, standards-window evidence, and routed supplements as provenance-bearing evidence. Source material is not treated as manuscript text; it is bound into the paper only through claim lanes, receipts, validators, normal forms, calibration rows, or explicit falsifier boundaries.

Expansion Rule For This Slot

Use past work to make lookups operational: lattice/forge/CAM address, invariant split, callable tool, and receipt boundary.

Primary Evidence Inputs

Source	Placement reason
08_Lattice_Closure.md	primary assigned source for the paper's formal spine
supplements/08_INTERNAL_REASONING_SUPPLEMENT.md	paper-local reasoning supplement contributing to definitions, proof sketch, and limitations

Secondary And Orbital Evidence Inputs

Source	Placement reason
supplements/SPECTRE_TILING_SUPPLEMENT.md	cross-cutting supplement; paper-relevant rows, receipts, and guard language are bound through evidence lanes
supplements/RECEIPT_VERIFIER_CATALOG.md	cross-cutting supplement; paper-relevant rows, receipts, and guard language are bound through evidence lanes
supplements/LATTICE_FORGE_UNIFICATION_SUPPLEMENT.md	cross-cutting supplement; paper-relevant rows, receipts, and guard language are bound through evidence lanes
supplements/INTERNAL_REASONING_PYRAMID_INDEX.md	cross-cutting supplement; paper-relevant rows, receipts, and guard language are bound through evidence lanes
supplements/INTERNAL_REASONING_5P_WINDOW_SUPPLEMENT.md	cross-cutting supplement; paper-relevant rows, receipts, and guard language are bound through evidence lanes
supplements/INTERNAL_REASONING_7P_WINDOW_SUPPLEMENT.md	cross-cutting supplement; paper-relevant rows, receipts, and guard language are bound through evidence lanes
supplements/INTERNAL_REASONING_9P_WINDOW_SUPPLEMENT.md	cross-cutting supplement; paper-relevant rows, receipts, and guard language are bound through evidence lanes
CAM_CLAIM_100_SCORE_AUDIT.md	series-wide audit/score/closure evidence; import paper-specific rows and leave global context outside the body
NSIT_TOOL_CLOSURE_MATRIX.md	series-wide audit/score/closure evidence; import paper-specific rows and leave global context outside the body
NSIT_REASONED_CLOSURE_PASS.md	series-wide audit/score/closure evidence; import paper-specific rows and leave global context outside the body
ORBITAL_DATA_CROSS_POPULATION_AUDIT.md	series-wide audit/score/closure evidence; import paper-specific rows and leave global context outside the body

Source	Placement reason
AGENT_CRYSTAL_WORK_HARVEST.md	series-wide audit/score/closure evidence; import paper-specific rows and leave global context outside the body
Additional source routes	Complete routing is maintained in the source-placement appendix

Crystal And Standards Evidence To Bind

- Paper Reworks crystal projection: 33 paper markdown files, 9 supplements, 5 queues, 6 lattice-forge unification artifacts, 140 faces, 140 vignettes, and 280 CAM nodes.
- Standards-window intake: 192/192 corpus conformance verdicts PASS; 140/142 obligations have candidate routes; 2/4/8/16/32 window lattice is available for cross-paper reads.
- Agent/crystal harvest: agent-generated memory, runtime standards starter, current runtime build, repo-harvest CAM, notebook-derived bundles, and downloaded crystal payloads are orbital evidence surfaces.
- NSIT inventory baseline: 220 validators, 1786 receipt/data artifacts, 1137 formal-paper-like artifacts, 114 supplements, and 86 memory/CAM artifacts.

Paper-Specific CAM/Score Rows

08	91	system-composition closeable	finite E8/E8x3/Niemeier chain, no-cost Leech lookup, lattice forge surfaces	expanded Leech invariants and nontrivial glue cosets
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Paper-Specific NSIT Closure Rows

PFC-2 E8 root enumeration	individual_tool_closure	E8Forge	Paper 08	e8_hemisphere_partition_receipt.json 5/5	240 roots, 120 antipodal pairs, clean split confirmed.
Lattice code chain 1->3->7->8->24->72	individual_tool_closure	lattice_forge/lattice_codes.py	Paper 08	lattice_code_chain_receipt.json 6/6	E8/Fano/Hamming/Golay/Leech chain closes.
O7 Niemeier:E8^3 partial carrier/Leech landing	partial_with_residue	enumerated_glue.py, nebe_gamma72.py	Paper 08	niemeier_leech_enumeration_receipt.json 6/6	E8^3 carriers and Leech landing close; exact glue reps were residue before later batch closure.
O2" F2 bridge algebraic core + unipotent orbits + substrate map	partial_with_residue	f2_majorana.py, unipotent_orbits.py, substrate_map.py	Paper 08	f2_bridge_unipotent_substrate_receipt.json 3/3	F2 Arf core, E8 unipotent orbit tables, substrate map close; registry population was separate.
O7 exact Niemeier:E8^3 glue cosets	batch_tool_unison_closure	E8Forge over three E8 blocks	Paper 08	o7_niemeier_e8cubed_glue_closed_receipt.json 7/7	Glue cosets = {0}; identity glue; 720 roots.
O2" registry population	batch_tool_unison_closure	f2_arf, lucas_recurrence, rule30_decomposition, f2_edge_glue validators	Paper 08	o2pp_registry_populated_receipt.json 6/6	314 facts accepted; 0 rejected.

Source Integration Summary

The legacy source and internal reasoning supplement are treated as source evidence rather than manuscript prose. Their contributions enter this paper as claim-lane entries, receipt or validator references, finite-domain statements, and limitation boundaries. Speculative or cross-domain interpretations remain separate from closed local results until an explicit adapter, citation, normal-form derivation, calibration row, or falsifier is attached.

Claim Binding Queue

This queue records the evidence discipline for claim strengthening. It separates claims that already have support from residues that remain in falsifier or open-obligation lanes.

Queue	Lane	Promote now	Bind next	Residue
Leech/E8/Golay Operational Lookup	receipt_bound_internal_result	Promote no-cost library lookup, code-chain, E8/Niemeier/Leech construction-surface claims when the lattice-forge API or receipt is named.	Attach exact lattice-forge API path, construction parameters, receipt JSON, and whether the claim is lookup, construction, or invariant-scope.	Expanded invariants, nontrivial glue-coset uniqueness, Gamma72, and physical interpretation remain separate dependencies.

Binding Discipline

Each queue item is represented as a delta:

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local claim at paper time
-> attached later source/tool/receipt/theorem
-> revised representation
-> invariant boundary and remaining residue

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This preserves the sequential story while allowing later tools, crystals, and standard formalisms to return through the proper lane.

Claim Delta Ledger

This ledger applies the paper's claim-binding queues as explicit back-application deltas. It keeps the sequential paper story intact while allowing later receipts, standard formalisms, CAM/crystal routes, and validators to sharpen the claim.

Delta	Local claim at paper time	Later evidence	Revised representation	Boundary kept
Leech E8 Golay Lookup	This paper may name lattice closure, E8/Niemeier/Leech surfaces, or terminal lattice addresses only at the scope stated locally.	Lattice-forge code-chain receipts, E8/Niemeier lookup machinery, Leech/Golay construction parameters, and related NSIT rows.	Promote operational lookup and construction-surface claims when the exact API/receipt path is attached.	Do not promote lookup into uniqueness, full invariant classification, Gamma72 closure, or physical interpretation without separate evidence.

The revised representation records the strongest claim supported by the current evidence package. Promotion into theorem or proposition language occurs only when the named evidence is attached in the manifest or workbook replay.

Source-Backed Evidence Summary

Routed archive and mirror cards are treated as provenance summaries rather than body text. They identify candidate receipts, validators, theorem registries, or supplemental source routes. A card strengthens this paper only when its contribution is bound to a claim lane, evidence object, and boundary.

Evidence card	Score	Publication contribution	Boundary
CQE-paper-08.25: Paper 8.25 - Toolkit for the Lattice Closure Template	81	Paper 8.25 describes the tools for reviewing the local lattice closure template. These tools expose the code/lattice checks and their audit boundaries; they do not close Leech or Gamma72 landing claims. The toolkit wo...	Source-backed support; promotion requires the paper-local claim lane and evidence boundary.
CQE-paper-08.50: Paper 8.50 - Lattice Closure Claim Contract	81	Paper 8.50 lets later papers import the lattice closure scaffold honestly. It keeps the verified local rungs useful while preserving global landings as separate proof obligations.	Source-backed support; promotion requires the paper-local claim lane and evidence boundary.
FORMAL: Paper 11 — C-Form: Theory Admission Gate Gluon	77	C = the admission filter Gluon — the trust anchor that filters external theories by Gluon mass matching. In the lattice_forge substrate, C is realized as the admission gate that: - Takes an external theory's G...	Source-backed support; promotion requires the paper-local claim lane and evidence boundary.
FINAL_FORMAL_PAPER: Complete Formal Claims: The Folded Strand	75	We present the complete closed-form claim set of the CQE_CMPLX corpus: - 33 papers = 33 folding operations on a self-complementary strand - 144 tools = cumulative analog kit = digital twin surface - 135 digital twins ...	Source-backed support; promotion requires the paper-local claim lane and evidence boundary.

Evidence card	Score	Publication contribution	Boundary
SUMMARY-V-32-Theorems-Registry: Summary Paper V — The 32 Theorems in One Table: Closed-Form Registry	75	This paper is the complete closed-form registry of all 32 theorems in the CQE_CMPLX corpus. Each theorem is stated precisely, given its formal context (where it is proven), and listed with its verifier. The table ...	Source-backed support; promotion requires the paper-local claim lane and evidence boundary.
Additional evidence cards	25 total	The complete card inventory is maintained in the archive evidence matrix.	Cards are source-discovery aids until bound to paper-local evidence.

Journal Apparatus

Article Type And Intended Venue Fit

This manuscript is written as a formal-methods and mathematical-systems paper inside the series **Formalizing LCR, Unifying Standard Models**. Its intended reader is a technical reviewer who needs claim boundaries, reproducibility routes, and cross-paper dependencies to be visible without relying on private context.

Structured Contribution Statement

Item	Statement
Publication ID	FLCR-09
Legacy source slot(s)	08
Ribbon role	order-1 slot-8: land into lattice, game, runtime, or terminal address surfaces
Proof form	terminal lookup, invariant split, and addressability proof
Received state	state emitted by prior slot 07 and reopened at original slot 08 (Lattice Closure)
Produced state	formal result of original slot 08 plus the same-family lift path toward slot 18

Claim-Evidence Table

Claim	Lane	Evidence to attach	Boundary
Theorem 9.1	cam_crystal_reapplication_result	Receipt, source card, validator, citation, or CAM route named in the paper manifest	The FLCR lattice ladder supports receipt-bound terminal lookup across the declared code/lattice chain.
Proposition 9.2	receipt_bound_internal_result	Receipt, source card, validator, citation, or CAM route named in the paper manifest	No-cost Leech lookup is a valid computational capability once the lookup receipt is attached, but it is not a proof of every Leech invariant.
Protocol 9.3	falsifier_or_open_obligation	Receipt, source card, validator, citation, or CAM route named in the paper manifest	Nontrivial glue, Gamma72, orbit, and physical-map claims require their own evidence lanes.

Figure Plan

Figure	Purpose	Caption
FLCR-09-F1	Terminal lookup address map	Schematic showing how FLCR-09 turns its received state into the produced state while preserving claim lanes and residue boundaries.
FLCR-09-F2	Evidence routing map	Diagram of source papers, archive cards, CAM/crystal routes, validators, and workbook replay paths used by this manuscript.
FLCR-09-F3	Same-family lift context	Diagram placing this paper in the original 00-79 ribbon family and showing predecessor/successor slots.

In-Text Figure FLCR-09-F1: Terminal lookup address map

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received state
-> Terminal lookup address map
-> formal claim lane
-> evidence object / receipt / citation
-> produced state
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-> remaining residue

Table Plan

Table	Purpose
FLCR-09-T1	Lookup/address table
FLCR-09-T2	Claim-lane/evidence-boundary table
FLCR-09-T3	Archive evidence card and source-backed upgrade table
FLCR-09-T4	Workbook replay and falsifier table

Worked Example

Resolve one CAM/lattice/forged address to a typed result and receipt boundary. The example must name the input object, the operation, the evidence object, the allowed revised claim, and the remaining boundary.

Nomenclature And Glossary

Term	Paper-local meaning
CAM	Defined by this paper as part of the <code>terminal_action</code> proof role; final definition is recorded in the paper-local glossary.
claim lane	Defined by this paper as part of the <code>terminal_action</code> proof role; final definition is recorded in the paper-local glossary.
crystal	Defined by this paper as part of the <code>terminal_action</code> proof role; final definition is recorded in the paper-local glossary.
lattice surface	Defined by this paper as part of the <code>terminal_action</code> proof role; final definition is recorded in the paper-local glossary.
lookup	Defined by this paper as part of the <code>terminal_action</code> proof role; final definition is recorded in the paper-local glossary.
query cache	Defined by this paper as part of the <code>terminal_action</code> proof role; final definition is recorded in the paper-local glossary.
receipt	Defined by this paper as part of the <code>terminal_action</code> proof role; final definition is recorded in the paper-local glossary.
terminal address	Defined by this paper as part of the <code>terminal_action</code> proof role; final definition is recorded in the paper-local glossary.

Appendix FLCR-09-A: Evidence Package

The appendix records:

1. Legacy source excerpts or source-card references used in the final claim ledger.
2. Receipt and verifier paths, including pass/fail/partial status.
3. CAM/crystal query or projection rows used by the paper, with source identity and boundary.
4. Standards-window and NSIT evidence used for conformance or routing.
5. Falsifier, calibration, or external-data rows that remain unresolved.

Data And Code Availability

The publication package contains the canonical Markdown sources, manifests, source-routing matrices, validation reports, and generated PDF/TeX outputs.

Code and verifier references are cited through manifests, archive evidence cards, receipt catalogs, or workbook replay tables. External datasets or standards citations must be attached before any calibration-bound claim is promoted.

Reviewer Checklist

Check	Reviewer question
Claim lane	Does every strong claim declare its lane and evidence type?
Boundary	Does the paper preserve what it does not prove?
Reproducibility	Is there a receipt, verifier, source card, citation, or replay table for the claim?

Check	Reviewer question
Cross-paper import	Does the paper cite the prior FLCR/OPR slot before consuming its result?
Interpretation	Does the analog layer preserve the addressed state while changing labels/access paths?

Declarations

No human-subjects, clinical, or animal-research protocol is asserted by this paper. External empirical claims require separate dataset, calibration, or domain-validation attachments. Competing-interest and funding statements are recorded in the submission metadata.

11. Publication Integration Addendum

11.1 Role In The Series

FLCR-09 (Lattice Closure And Terminal Addresses) occupies the `terminal_action` role at lift depth 0. The paper receives state emitted by prior slot 07 and reopened at original slot 08 (Lattice Closure). It produces formal result of original slot 08 plus the same-family lift path toward slot 18. The state transition is: land the state in a terminal lattice, game, runtime, or address surface.

The paper-local proof form is:

`terminal lookup, invariant split, and addressability proof`

The integration result is an addressable lookup surface with invariant and residue split. This addendum records how that result is consumed by the publication series without relying on editorial instructions or private working context.

11.2 Formal Carrier

Formal carrier	Function
lattice theory	Formal carrier for the paper-local result.
finite-state systems	Formal carrier for the paper-local result.
terminal object/addressing templates	Formal carrier for the paper-local result.
lookup-table construction and invariant checks	Formal carrier for the paper-local result.

The LCR, CAM, crystal, forge, and analog vocabulary functions as a typed access layer over these carriers. A relabeling is admissible only when the addressed object, evidence lane, boundary, and downstream import rule remain attached.

11.3 Claim Ledger

Claim	Lane	Statement
Theorem 9.1	<code>cam_crystal_reapplication_result</code>	The FLCR lattice ladder supports receipt-bound terminal lookup across the declared code/lattice chain.
Proposition 9.2	<code>receipt_bound_internal_result</code>	No-cost Leech lookup is a valid computational capability once the lookup receipt is attached, but it is not a proof of every Leech invariant.
Protocol 9.3	<code>falsifier_or_open_obligation</code>	Nontrivial glue, Gamma72, orbit, and physical-map claims require their own evidence lanes.

These claims are consumed at their stated lane. A stronger downstream statement creates a new claim envelope with its own evidence object.

11.4 Evidence Package

Evidence class	Routed items	Status
Legacy sources	<code>08_Lattice_Closure.md</code> , <code>supplements/08_INTERNAL_REASONING_SUPPLEMENT.md</code>	routed evidence
Evidence inputs	<code>CAM_CLAIM_100_SCORE_AUDIT.md</code> , <code>NSIT_TOOL_CLOSURE_MATRIX.md</code> , <code>NSIT_REASONED_CLOSURE_PASS.md</code> , <code>PAPER_REWORKS_CRYSTAL_PROJECTION.json</code>	routed evidence

Evidence class	Routed items	Status
CAM/crystal sources	PAPER_REWORKS_CRYSTAL_PROJECTION.json, AGENT_CRYSTAL_WORK_HARVEST.json, runtime standards artifact, notebook-derived source bundle	routed evidence
Standards-window inputs	window_summary.json, node DBs, window DBs	routed evidence
Receipts	external requirement	not attached in this source package
Validators	external requirement	not attached in this source package

Standards-window, runtime, and notebook-derived artifacts are treated as evidence-routing surfaces. They support source discovery, conformance checking, and reapplication, but they do not replace citations, receipts, normal-form derivations, calibration rows, or falsifiers.

11.5 Results Representation

The paper's results section is organized around five publication objects:

1. the exact object acted on at this slot;
2. the admissible operation or transformation;
3. the receipt, enumeration, derivation, lookup, or external theorem supporting the operation;
4. the residue or boundary left after the result;
5. the downstream import rule.

11.6 Figures And Tables

Figure	Role
FLCR-09-F1: Terminal lookup address map	Visualizes the proof object, routing, or boundary.
FLCR-09-F2: Evidence routing map	Visualizes the proof object, routing, or boundary.
FLCR-09-F3: Same-family lift context	Visualizes the proof object, routing, or boundary.
Table	Role
FLCR-09-T1: Lookup/address table	Records claim lanes, evidence, sources, or falsifiers.
FLCR-09-T2: Claim-lane/evidence-boundary table	Records claim lanes, evidence, sources, or falsifiers.
FLCR-09-T3: Archive evidence card and source-backed upgrade table	Records claim lanes, evidence, sources, or falsifiers.
FLCR-09-T4: Workbook replay and falsifier table	Records claim lanes, evidence, sources, or falsifiers.

11.7 Limits And Falsifiers

The paper proves only the object and domain declared in its claim ledger. Physical, Standard Model, observer, calibration, or synthesis claims require the additional lane evidence stated by their destination papers. CAM/crystal and forge language is operational only when represented as an address, lookup, receipt, validator, or reapplication path.